

MD Saib Hossain

AI Researcher | Machine Learning Engineer Dhaka, Bangladesh |

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<https://github.com/Saibhossain>

Profile

AI Engineer specializing in Agentic AI, LLM Applications, Retrieval-Augmented Generation (RAG), and AI-Powered Business Automation. Experienced in designing and deploying production-ready AI pipelines using LangSmith, LangGraph, FastAPI, and DeepAgents workflows. Skilled in fine-tuning (LoRA/QLoRA), scalable AI products through RAG, workflow orchestration, evaluation frameworks, and API-driven architectures. Strong background in stakeholder collaboration, technical problem-solving, and end-to-end AI system development from prototype to deployment.

Experience

1. Research Assistant | Institute of Advanced Research (IAR) - (UIU) | Nov 2025 - Present

Deep learning pipelines for medical image analysis. Extracted & analyzed radiomic features supported model evaluation and statistical correlation studies. Documented experiments and findings of model reliability, data bias, and clinical applicability, relevant to healthcare AI governance. Applied algorithms, data structures, and optimization techniques to improve search efficiency and workflow latency.

2. AI Engineer | FireAI (Betopia Group) | April 2026 - May 2026

I'll be focusing on building the next generation of agentic AI to completely automate complex multi-step tasks and architecting multi-agent systems for production. Multi-agent AI systems to automate multi-step business workflows with Python and LLM orchestration for production.

3. Machine Learning Engineer Intern | Fluvo Soft (Remote) | Nov 2025 - Mar 2026

Designed a synthetic face generation pipeline with StyleGAN2 and Stable Diffusion XL (SDXL) to address data scarcity, and also developed an Agentic RAG system for the customer support and hr team workflow automation.

4. Volunteer Research Assistant | Ruach Tech (London, UK- Remote) Jan - Mar 2026

Conducted research on AI safety and alignment, analyzed and summarized advanced AI research papers. Supported the development of safe AI system frameworks for ongoing R&D projects.

Projects

1. ScholarAgent: AI Multi-Agent Scholarship Advisor

Built a multi-agent scholarship intelligence system using CrewAI and Ollama with (34–42s) per-agent inference latency and autonomous academic planning workflows. A CrewAI-based agentic AI pipeline for scholarship retrieval, roadmap generation, and SOP automation with real-time web search and (146s) end-to-end execution latency. <https://github.com/Saibhossain/ScholarAgent.git>

2. Self-Reflective Retrieval-Augmented Generation (Self-RAG):

Built an autonomous RAG system with query rewriting, retrieval relevance grading, hallucination detection, and answer validation using LangGraph, LangSmith, and OpenAI frontier models to improve factual accuracy and response reliability. https://github.com/Saibhossain/self_rag_system.git

3. Multi-Agent Image Generation Pipeline (CrewAI + OpenAI + Langtrace)

Architected a modular multi-agent creative generation system with separated responsibilities across prompt engineering, generation, and automated critique stages. Implemented production-style observability, tracing, and cost instrumentation for agent workflows using Langtrace. Built reusable tool interfaces for future extensions to multiple image providers and feedback loops. \$0.0076 per workflow. https://github.com/Saibhossain/generate_image_with_crewai.git

4. Agentic AI Technical Interviewer

Engineered a non-linear, autonomous AI interviewing agent that dynamically parses PDF resumes to generate tailored technical assessments. Orchestrated complex, stateful conversation flows including real-time answer evaluation, clarifying hints, and adaptive follow-ups.

<https://github.com/Saibhossain/Interview-agent.git>

5. MedAssist-LLM-iPhone

Fine-tuned a lightweight **Qwen2.5-0.5B-Instruct LLM** on a medical Question-Answering dataset and optimized it for fully offline inference on an **iPhone X**. <https://github.com/Saibhossain/MedAssist-LLM-iPhone.git>

6. Real-Time Vision + LLM Agent “LLM with Eyes.”

Developed an autonomous vision agent using YOLOv11 and the Gemini 2.5 Flash LLM. Pipeline segments objects from a live camera feed in real-time and routes visual data to an LLM for context-aware identification.

<https://github.com/Saibhossain/Eye-of-LLM.git>

7. Educational ChatBot. <https://phi-chat-ui.vercel.app/>

Publications (First Author)

IEEE ICCIT 2025 (3 Papers):

- Fracture Classification (ResNet, YOLO, DeepLabV3) — AUC +3.8%, mAP +4.2% [Link](#)
- Bangla Sign Language System — 98%+ accuracy <https://ieeexplore.ieee.org/document/11491587>
- Satellite-based Environmental Analysis <https://ieeexplore.ieee.org/document/11490302>

Leadership & social work

Volunteer / Social Worker | Alo Bhubon Trust (2025 – Present) [Link](#)

Contributed to community development and outreach programs. Supported initiatives in health, education, and climate awareness

General Secretary | National Children’s Task Force (NCTF) Satkhira 2014 – 2017 [Link](#)

Led district-level coordination and stakeholder collaboration. Managed community initiatives and policy monitoring programs

Achievements & Awards

1st Runner-up – Kaggle Competition (Deep Learning in Computational and Spatial Biology)

Best Abstract Award, same competition (Spatial Biology)

Technical Skills

Programming & Frameworks: C++, Python, PyTorch, TensorFlow, Scikit-learn, OpenCV

LLM & RAG Systems: OpenAI & Anthropic API Integration, Self RAG, Agentic RAG, FAISS, Qdrant,

Agentic AI Tech: LangChain, LangGraph, CrewAI, DeepAgent, Langsmith.

Computer Vision: Object detection (YOLO), segmentation, medical imaging, dataset curation

Algorithms & Optimization: Data Structures, Latency Optimization, System Design.

Development: AI Product Prototyping & MVP Development, Technical Requirement Analysis

Tools: Github, Ollama, Docker, Streamlit, Supabase

Education

United International University, DHAKA – BSCSE

February 2022 – 2026 - **CGPA: 3.45**

Additional Strengths

- Strong scientific writing and research presentation skills
- Strategic planning and communication

Certifications

Python and SQL - [Link](#),

Pytorch - [Link](#)

IBM AI Engineering - [Link](#)

MCP-Advanced Topics- [Link](#)

AWS AI Practitioner Challenge-[link](#)

Deep Agents - [Link](#)